



DAA.

Thứ ngày . .

Bài tập 1.

1.

$$2.4.1a) x(n) = 2x(n-3) \text{ for } n > 1, x(1) = 1.$$

$$\begin{aligned} \rightarrow x(n) &= 2x(n-3) \\ &= 2[2x(n-6)] = 2^2 x(n-6) \\ &= 2^2 [2x(n-9)] = 2^3 x(n-9) \\ &= \dots \\ &= 2^{\frac{n-1}{3}} x(n - (n-1)) \\ &= 2^{\frac{n-1}{3}} x(1) = 2^{\frac{n-1}{3}} \end{aligned}$$

$$2.4.1b) x(n) = x(n-2) - 2 \text{ for } n > 1, x(1) = 0.$$

$$\begin{aligned} \rightarrow x(n) &= x(n-2) - 2 \\ &= x[n(n-4) - 2] - 2 = x(n-4) - (2+2) \\ &= [x(n-6) - 2] - (2+2) = x(n-6) - (2+2+2) \\ &= \dots \\ &= [x(n - (n-1)) - 2] - \underbrace{(2+2+\dots+2)}_{\frac{n-1}{2}} \\ &= 0 - \underbrace{(2+\dots+2)}_{\frac{n-1}{2}} \\ &= 0 - 2 \frac{n-1}{2} = -n+1. \end{aligned}$$

$$2.4.1c) x(n) = 2x(n-2) - n \text{ for } n > 0, x(0) = 0$$

$$\begin{aligned} \rightarrow x(n) &= 2x(n-2) - n \\ &= 2[2x(n-4) - (n-2)] - n \\ &= 2^2 x(n-4) - 2(n-2) - n \\ &= \dots \\ &= 2^{n/2} x(n-n) - [2^{\frac{n}{2}-1} (n - (n-2)) + \dots + n] \\ &= 2^{n/2} x(0) - (n + 2(n-2) + \dots + 2^{n/2}) \\ &= -\underbrace{(n + 2(n-2) + \dots + 2^{n/2})}_{n/2} \end{aligned}$$

2.4.2.

* Setup:
$$F(n) = \begin{cases} 1 & n \leq 1 \\ F(\frac{n}{2}) + 1 & n > 1 \end{cases}$$

* Solve: Đặt $n = 2^m$

$$\begin{aligned} F(n) &= F(2^m) = F(2^{m-1}) + 1 \\ &= F(2^{m-2}) + 1 + 1 \\ &= \dots \\ &= F(2^{m-m}) + \underbrace{(1 + \dots + 1)}_m \\ &= F(1) + m \\ &= 1 + \log_2 n = O(\log(n)). \end{aligned}$$

2.4.8.

a) def power-of-two(n):

if $n == 0$:

return 1

else:

half-power = power-of-two($n-1$).

return half-power + half-power.

b) Gọi $A(n)$ là số phép cộng tăng thêm 2^n .

Base case: $A(0) = 0$

Recursive case: $A(n) = 2A(n-1) + 1$

Thử với các số nhỏ:

$$A(1) = 2A(0) + 1 = 2 \cdot 0 + 1 = 1.$$

$$A(2) = 2A(1) + 1 = 2(1) + 1 = 3.$$

$$A(3) = 2A(2) + 1 = 2(3) + 1 = 7.$$

$$\Rightarrow A(n) = 2^n - 1.$$

C/m bằng quy nạp:

- Base case: $A(0) = 2^0 - 1 = 0$ (luôn đúng).

- Induction case: Giả sử $A(k) = 2^k - 1$ đúng với mọi k .

$$\text{Với } k+1: A(k+1) = 2A(k) + 1 = 2(2^k - 1) + 1 = 2^{k+1} - 2 + 1 = 2^{k+1} - 1.$$



2.

a) $T(1) = 1$, and for all $n \geq 2$, $T(n) = 3T(n-1) + 2$.

$$\rightarrow T(n) = 3T(n-1) + 2.$$

$$= 3(3T(n-2) + 2) + 2.$$

$$= 3^2(T(n-2)) + (2+2+2).$$

$$= \dots$$

$$= 3^{n-1}T(1) + \underbrace{(2+\dots+2)}_{n-1}.$$

$$= 3^{n-1} + 2 \cdot (n-1).$$

$$= O(3^n).$$

b) $T(1) = 1$, and for all $n \geq 2$, $T(n) = 2T(n-1) + n - 1$.

$$\rightarrow T(n) = 2T(n-1) + n - 1.$$

$$= 2(2T(n-2) + n-1) + n-1.$$

$$= 2^2(T(n-2)) + 2(n-2) + n-1.$$

$$= 2^3(T(n-3)) + 4(n-3) + 2(n-2) + (n-1).$$

$$= \dots$$

$$= 2^k T(n-k) + \sum_{i=0}^{k-1} 2^i (n-i-1).$$

Chọn $k = n-1$, khi đó:

$$T(1) = 1 \Rightarrow T(n) = 2^{n-1}T(1) + \sum_{i=0}^{n-2} 2^i (n-i-1).$$

$$= 2^{n-1} + \sum_{i=0}^{n-2} 2^i (n-1-i).$$

$$= 2^{n-1} + n \sum_{i=0}^{n-2} 2^i - \sum_{i=0}^{n-2} i 2^i - \sum_{i=0}^{n-2} 2^i.$$

$$= 2^{n-1} + n(2^{n-1} - 1) - ((n-3)2^{n-1} + 2) - (2^{n-1} - 1).$$

$$= 2^{n-1} + 2^{n-1}n - n + (n-5)2^{n-1} - 2 - 2^{n-1} + 1.$$

$$= 2^{n-1} + (n-3)2^{n-1} + 2^{n-1}n - 2^{n-1} - n - 2 + 1.$$

$$= 2^{n-1} + (n-3)2^{n-1} + 2^{n-1}n - 2^{n-1} - n - 1.$$

$$= 2^{n-1} + (n-3+n-1)2^{n-1} - n - 1.$$

$$= 2^{n-1} + (2n-4)2^{n-1} - n - 1.$$

$$= 2^{n-1}(2n-3) - n - 1.$$

$$= O(2^n \cdot n).$$

e) $T(1) = 4$, and for all $n \geq 2$ a power of 2, $T(n) = 2T\left(\frac{n}{2}\right) + 3n + 2$.

$$T(n) = 2T\left(\frac{n}{2}\right) + 3n + 2. \quad a = 2, b = 2, d = 1.$$

Sử dụng ^{Master} General Theorem với $a = b^d \Rightarrow T(n) = O(n \log n)$.

b) $T(1) = 3$, and for all $n \geq 2$ a power of 3, $T(n) = 4T\left(\frac{n}{3}\right) + 2n - 1$.

$$T(n) = 4T\left(\frac{n}{3}\right) + 2n - 1.$$

Đặt $n = 3^m$.

$$\text{Ta có: } T(n) = T(3^m) = 4T(3^{m-1}) + 2 \cdot 3^m - 1. \quad (1)$$

$$= 4(4T(3^{m-2}) + 2 \cdot 3^{m-1} - 1) + 2 \cdot 3^m - 1.$$

$$= 4^2 T(3^{m-2}) + 8 \cdot 3^{m-1} - 4 + 2 \cdot 3^m - 1.$$

$$= 4^2 T(3^{m-2}) + (9-1) \cdot 3^{m-1} - 4 + 2 \cdot 3^m - 1.$$

$$= 2 \cdot 4^m + 2(9^m - 3^{m+1}) - \frac{4^m - 1}{3}.$$

$$\Rightarrow T(n) = O(n \log_3 4).$$

Bài tập 2: